



From DOT Data Complexity to Trusted Decisions

A practical foundation for data, GIS, analytics, and AI.

Justin O'Daniel · Databricks, State Government

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DOTs Don't Have a Data Shortage

They have a data-to-decision gap.

01

**No enterprise
view**

02

**Low trust in
the data**

03

**Funding
pressure**

04

**Reactive
operations**

05

**Reporting
burden**

Safety, mobility, and cost problems — not IT problems.



Why That Gap Is Hard to Close

The same problems, from the engineering side.

01 A copy for every question

02 Pipelines maintained by hand

03 No shared definitions or lineage

04 Built for batch reporting

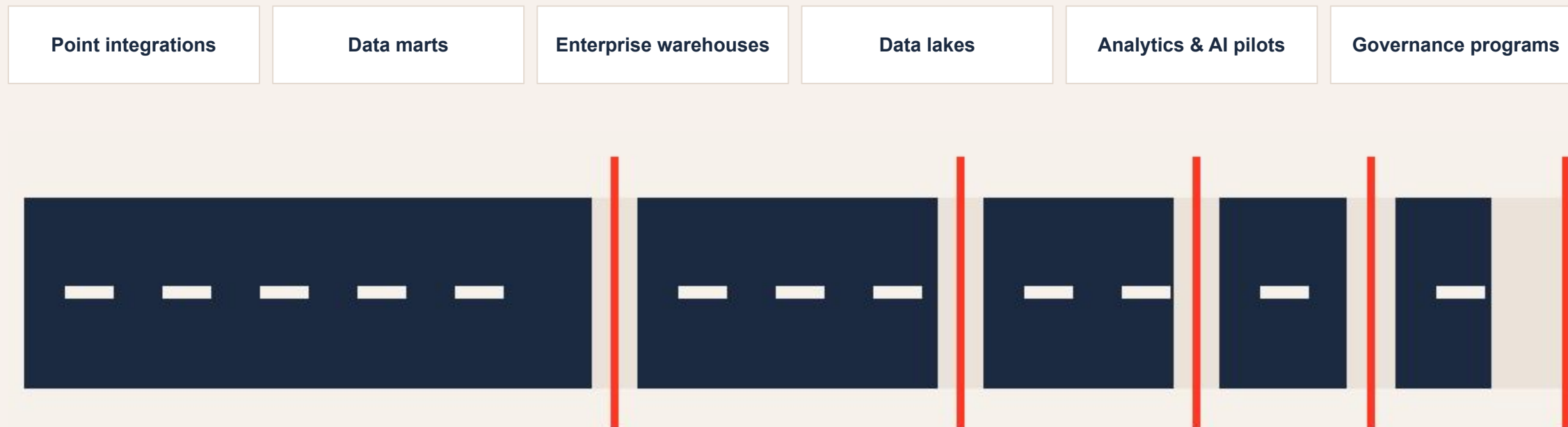
05 Governance lives in documents

06 Specialists rebuilding the same work

Every one is a tax on the same scarce capacity — and it compounds with each new request.



It Works Locally. Then It Stalls.



Every new use case rebuilds the same integration, quality, security, access, and operationalization work.

None of these failed. They created real, local value. They just do not compound.

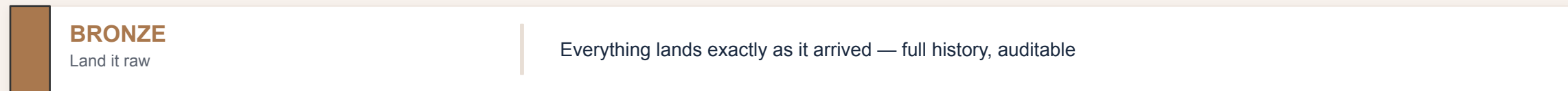
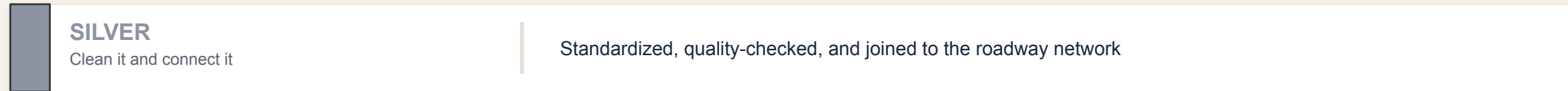
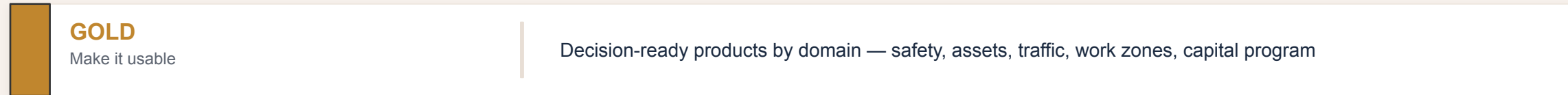


HOW IT WORKS

Land It Raw. Clean It. Make It Usable.

One path for every source — delivered back into the applications your teams already use.

PROJECTED INTO THE APPLICATIONS DOTs ALREADY USE



THE SYSTEMS DOTs ALREADY RUN



Plus bridge and pavement management, finance, weather, documents, and third-party data.



WHY THIS IS DIFFERENT

A Foundation, Not Another Data Project

THE USUAL PATH

ON A GOVERNED FOUNDATION

An extract for every consumer



One governed copy

A warehouse for structured reporting



Streaming, spatial, documents, and AI together

A lake you cannot fully trust



Quality and lineage built in

Pilots that end with the funding



Data products that outlive the project

Governance in a binder



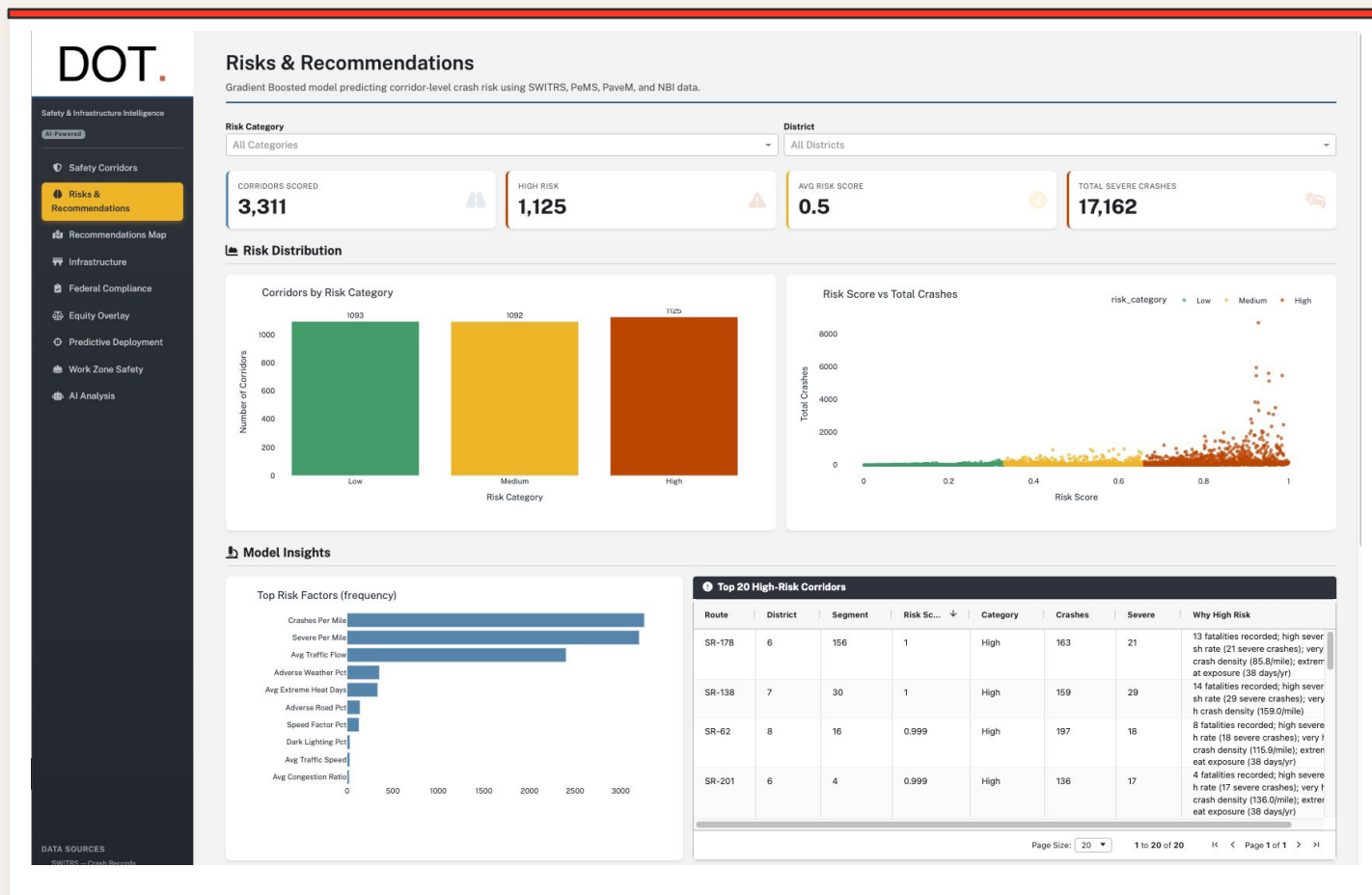
Governance enforced in the tools

Build it once. Every use case after the first one is cheaper.



1. One Governed Foundation. Many DOT Mission Outcomes.

The dashboard is visible — but the real value is the governed data, reuse, and mission breadth behind it.



Safety Corridors

Risks & Recommendations

Recommendations Map

Infrastructure

Federal Compliance

Equity Overlay

Predictive Deployment

Work Zone Safety

AI Analysis

Nine DOT domains

DATA SOURCES

SWITRS — Crash Records
FARS — Fatal Crash Data
PeMS — Traffic Flow & Speed
HPMS — Roadway Characteristics
PaveM — Pavement Condition
NBI — Bridge Inventory
CalEnviroScreen — Equity
Cal-Adapt — Climate Projections
Caltrans LCS — Live Work Zones

Sources combined once

Unify heterogeneous DOT data once

Crash, roadway, traffic, pavement, bridge, weather, work-zone, equity, project, and policy data.

Govern it as reusable data products

Standardize definitions, preserve lineage, control access, and make trusted information discoverable.

Use it across the DOT — not for one dashboard

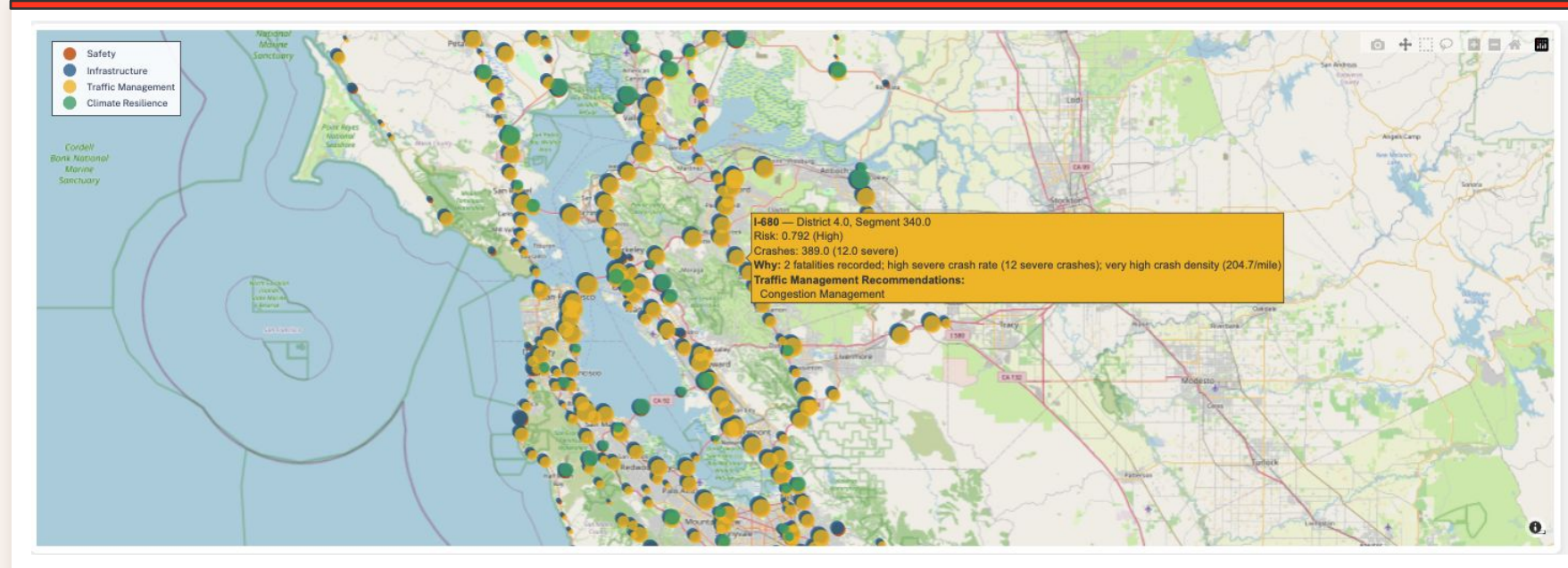
Safety, infrastructure, compliance, equity, operations, mapping, and AI all draw from the same foundation.



3. Put Risk, Location, and Recommended Actions on the Same Map

Location becomes the common language connecting evidence, policy, and field decisions.

DISTRICT VIEW



01

Move from statewide portfolio to district, corridor, and segment.

02

Connect a location to its risk evidence and contributing conditions.

03

Surface policy-informed countermeasures for DOT expert review.

SELECTED CORRIDOR

I-680 — District 4, Segment 340

High Risk Score: 0.792

Crashes: 389.0 Severe: 12.0 Killed: 2.0

Why this risk score:

2 fatalities recorded; high severe crash rate (12 severe crashes); very high crash density (204.7/mile)

Based on this risk profile, the following actions are recommended:

Safety

Comprehensive Safety Review

Proven Safety Countermeasure (Caltrans PSC)

Road Safety Audits (RSA): ~25% crash reduction (CMF source: FHWA Road Safety Audit Guidelines 2006; Elvik 2001 meta-analysis)

FHWA HSIP (90%) USDOT SS4A (80%) NHTSA 402 (80%)

- ☐ Commission Road Safety Audit (RSA) per FHWA RSA Guidelines — PSC proven to reduce crashes 20–40%
- ☒ Apply systemic low-cost countermeasures (signing, delineation, markings) across corridor network
- ☒ Request HSIP funding eligibility review — RSA findings support project justification
- ☐ Schedule field visit with District Traffic Safety and initiate Local Road Safety Plan (LRSP) if not active
- ☐ Review CA SHSP emphasis areas applicable to this corridor's crash pattern

Severe Crash Countermeasures

Proven Safety Countermeasure (Caltrans PSC)

Median Barrier: ~25% crash reduction (CMF source: FHWA CMF Clearinghouse #749; NCHRP Report 711)

FHWA HSIP (90%) USDOT SS4A (80%) NHTSA 402 (80%)

- ☐ Evaluate median barrier installation (CMF 0.25 for crossover crashes, 75% reduction) — HSIP-eligible
- ☐ Assess roundabout feasibility at key intersections (CMF 0.37 for fatal/serious, 63% reduction)
- ☒ Consider Reduced Left-Turn Conflict design (J-turn/MUT) where applicable (CMF 0.51 for serious injury)
- ☒ Improve roadside design at curves: remove/shield fixed objects per AASHTO Roadside Design Guide
- ☐ Submit HSIP application for Proven Safety Countermeasures targeting severe and fatal crash types

Infrastructure



4. Turn a Priority Corridor into a Fundable, Accountable Action Plan

Connect evidence to countermeasures, potential funding programs, and the next workflow step.

STEP 1 · EVIDENCE TO COUNTERMEASURES AND FUNDING

Federal Funding Opportunities 8 programs across FHWA, NHTSA, USDOT
Up to 90% Federal Share 6 Formula 3 Competitive Est. federal offset: \$6M - \$57M of \$7M - \$64M total

Route SR-178 — District 6.0 — Segment 156.0 Est. \$2M - \$17M 8 grants 1,000

Severe Crash Countermeasures Severe Per Mile \$500K - \$5M
Install median barriers, rumble strips, and cable median barriers; review sight distances

Eligible Federal Grants

- Highway Safety Improvement Program** ALN 20.205
90% Federal Formula \$3.2B nationally
Largest dedicated safety program. Funds infrastructure countermeasures on all public roads consistent with the state SHSP.
- Safe Streets and Roads for All** ALN 20.939
80% Federal Competitive \$1B nationally
Competitive grants for comprehensive safety action plans and implementation projects. Awards \$100K-\$25M per project.
- State & Community Highway Safety (Section 402)** 23 CFR §1300.11
80% Federal Formula \$297M nationally
Triennial HSP formula grants. Requires performance targets (C-1 fatalities, C-2 serious injuries, C-3 fatality rate/VMT), data-driven problem identification, and evidence-b
- Occupant Protection (Section 405b)** 23 CFR §1300.21
80% Federal Formula \$30M nationally
Occupant protection grants for seat belt enforcement and child restraint programs. Requires primary seat belt law, high-visibility enforcement plan, and community educi



STEP 2 · EXPERT REVIEW AND SELECTION

Traffic Management

Congestion Management

Proven Safety Countermeasure (Caltrans PSC)
Corridor Access Management: ~35% crash reduction (CMF source: FHWA Access Management Manual; CMF Clearinghouse #4021)

FHWA CMAQ (80%) FHWA NHPP (80%) USDOT SMART (60%)

- ☐ Apply Corridor Access Management to reduce conflict points and improve flow (CMF 0.65)
- ☐ Evaluate Variable Speed Limit system on congested corridors (CMF 0.85 total crashes)
- ☐ Request TMS signal timing optimization and adaptive control evaluation
- ☐ Evaluate ramp metering adjustments — reduces merge conflicts and rear-end crashes
- ☐ Submit Congestion Management Program project nomination

Submit Selected Actions

24 actions available — check the ones to initiate



STEP 3 · ROUTED INTO DOT WORKFLOW

Actions Submitted

Actions for I-680 — District 4, Segment 340 have been successfully submitted.

District coordinators will be notified and work orders will be initiated for all selected items.

OK

Connect risk evidence to relevant funding pathways.

Give safety, planning, and grant teams a defensible starting point.

Move selected actions into accountable DOT workflows.



5. Move from Reporting Yesterday's Risk to Prioritizing Today's Operations

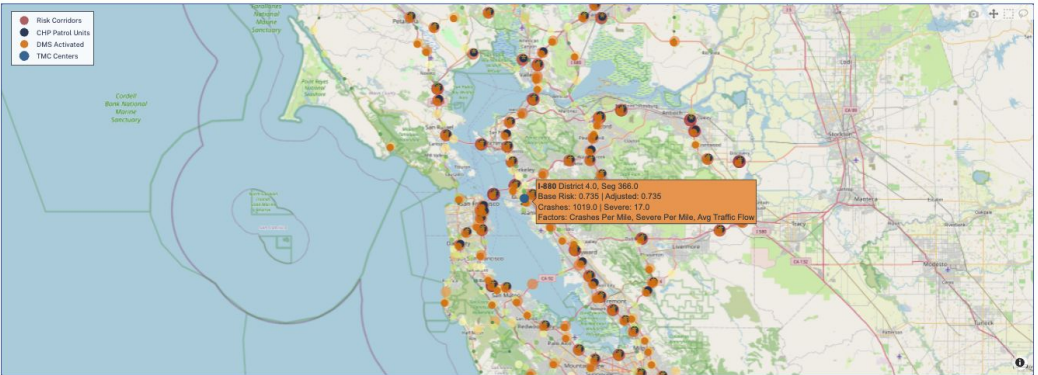
Use time, location, and resource context to focus patrol, message-sign, and traffic-management actions.

TIME-OF-DAY SCENARIO

Select a time period to see how predictive deployment shifts across the day.

- ☐ Late Night (12-5 AM)
- ☐ Early Morning (5-7 AM)
- ☐ Morning Commute (7-9 AM)
- ☒ Midday (9 AM-3 PM)
- ☐ Evening Commute (3-7 PM)
- ☐ Evening (7-10 PM)
- ☐ Night (10 PM-12 AM)

DEPLOYMENT MAP



RECOMMENDED ACTION PACKAGES · ACCEPTED BY AN OPERATOR

📍 Route SR-178 — District 6 — Seg 156 Risk: 1.000

~28 crashes prevented (4 severe)

CHP Patrol Peak hours (3-7 PM)
Deploy patrol unit to SR-178 District 6

DMS Warning Triggered by real-time risk threshold
Activate 'HIGH CRASH RISK — REDUCE SPEED' on SR-178

TMC Alert Continuous monitoring
District 6 TMC — monitor SR-178 Seg 156

Accept & Deploy

📍 Route SR-138 — District 7 — Seg 30 Risk: 1.000

~27 crashes prevented (5 severe)

CHP Patrol Peak hours (3-7 PM)
Deploy patrol unit to SR-138 District 7

DMS Warning Triggered by real-time risk threshold
Activate 'HIGH CRASH RISK — REDUCE SPEED' on SR-138

TMC Alert Continuous monitoring
District 7 TMC — monitor SR-138 Seg 30

Accept & Deploy

Risk changes by location and time of day.

Compare elevated-risk corridors with available operational resources.

Prioritize action packages for human review and execution.

MODEL-ESTIMATED POTENTIAL IMPACT · ILLUSTRATIVE AND SUBJECT TO DOT VALIDATION

Projected Crash Reduction

Based on Waycare I-15 pilot results (17% **primary crash reduction**), predictive deployment on these 83 corridors could prevent an estimated **8,794 crashes** annually.
Estimated **280 severe crashes prevented** across 83 highest-risk corridors during *Midday (9 AM-3 PM)* conditions.

Crashes on Deployed Corridors
51,732

Severe Crashes
1,647

Projected Prevented
8,794

Severe Prevented
280



6. Start Now: Ask Governed DOT Data Questions in Plain English

Genie provides a practical on-ramp to trusted data and AI — before a DOT builds its next custom application.

AI Analysis

Natural language data exploration powered by Databricks Genie.

Genie Space AI Chat

Ask questions about Caltrans data in plain English. Powered by Databricks Genie.

- What are the top 10 corridors by total crashes?
- Which corridors have the highest severe crash rate per mile?
- How many bridges are in each condition rating category?
- What is the average pavement PCI by Caltrans district?
- What are the top 15 primary collision factors by crash count, excluding 'Other'?
- Which corridors have the highest percentage of nighttime crashes?
- Which districts have the most corridors with poor pavement and high crash rates?

How many bridges are in each condition rating category?

There are 12,523 bridges rated as Fair, 11,796 bridges rated as Good, and 1,527 bridges rated as Poor. The majority of bridges fall into the Fair and Good categories, with significantly fewer in Poor condition.

What is the average pavement PCI by Caltrans district?

The average pavement PCI by Caltrans district ranges from 58.07 (District 2) to 60.91 (District 9). Notable data points include:

- District 9: 60.91
- District 12: 60.85
- District 7: 60.55
- District 6: 60.16
- District 1: 60.03

District 9 has the highest average PCI, while District 2 has the lowest among the districts shown.

Which corridors have the highest severe crash rate per mile?

Some of the corridors with the highest severe crash rate per mile are:

- I-10 (District 7, Segment 72.0): 185.0 severe crashes per mile
- US-101 (District 7, Segment 134.0): 148.42 severe crashes per mile
- SR-134 (District 7, Segment 2.0): 132.0 severe crashes per mile
- SR-210 (District 8, Segment 6.0): 124.55 severe crashes per mile
- SR-91 (District 12, Segment 34.0): 116.32 severe crashes per mile

01

Ask in plain English against approved DOT data

Questions run against the governed tables the DOT has already defined and secured.

02

No new report, extract, or custom dashboard

Trusted answers become available without another request queue or one-off build.

03

Start with high-value questions today

Expand into richer workflows and applications over time, on the same foundation.

QUESTIONS A DOT TEAM COULD ASK ON DAY ONE

Which corridors have the highest severe-crash rate per mile?

Which districts combine poor pavement condition with elevated crash risk?

How many bridges fall into each condition rating?

Which routes show the highest severe crashes during the morning commute?



A REALISTIC PATH

Start with One DOT Priority. Build for Many.

A DOT does not need a big-bang modernization program to get value from a foundation.

PHASE 1

Start with governed self-service

Point Genie at one trusted dataset and start answering questions people are already waiting on.

EXAMPLES Safety corridors · pavement and bridge condition · federal reporting pulls · the report that eats a person-week every month

PHASE 2

Establish reusable data products and governance patterns

Turn what the first use case produced into governed products the next team can pick up.

EXAMPLES Roadway network · assets · safety · traffic · projects · weather

PHASE 3

Scale across the DOT ecosystem

Extend the same foundation outward, including to partners who depend on your data.

EXAMPLES Operations · planning · maintenance · capital delivery · local partners · MPOs · public safety · advanced AI

Start with the mission outcome. Build a data foundation that gets more valuable with every use case.



DOT Use-Case Portfolio

Where a governed foundation typically pays for itself first — organized the way DOTs organize themselves.

Safety

Corridor and intersection prioritization · systemic risk screening · HSIP and SHSP analysis · vulnerable road user analysis

Asset management

Pavement and bridge condition · treatment timing · lifecycle cost · inspection backlog · TAMP support

Traffic operations and work zones

Congestion and reliability · incident response · work-zone impact and safety · signal and ramp performance

Capital program and project delivery

Portfolio visibility · schedule and cost variance · contractor performance · construction document analysis

Federal reporting and performance

HPMS, HSIP, NBI and MAP-21 measures · repeatable submissions · audit-ready lineage

Climate resilience and emergency response

Flood, heat, and storm exposure · vulnerable asset screening · event operations · recovery tracking

Data sharing and partner collaboration

MPOs and local agencies · public safety · transit and freight partners · open data without extract sprawl

AI and document intelligence

Inspection notes and reports · permits and contracts · imagery review · assistants grounded in governed data



Simplify the Data and AI Journey

01**Keep what works**

Keep the systems and GIS investments that deliver mission value.

02**Remove the friction**

Reduce fragile integrations, data copies, and manual preparation.

03**Govern it together**

Govern data, analytics, and AI in one place, not three.

04**Make it compound**

Turn one high-value DOT use case into a reusable foundation.

Databricks is not another data silo. It is the governed foundation that helps DOTs connect data, GIS, analytics, and AI — so teams can make safer, faster, more defensible decisions.

THE ASK Name one high-value DOT use case — and the reusable data products it would create.

